# Syncubator

## Design Documentation

A neonatal incubator with alarms, climate control and remote monitoring

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|---|---|
| Team | Syncubator (Team ID HSC\|HP\|00026) |
| Theme | Healthcare & Wellbeing (Hack for Social Cause) |
| Institution | Indian Institute of Technology Mandi, Kamand, Himachal Pradesh 175075 (AISHE code U-0184) |
| Mentor | Dr. Gajendra Singh, IIT Mandi |
| Status | Research prototype. Not a certified medical device and not for use on patients. |
| Version | October 2026 |

## 1. Purpose and scope

This document describes how Syncubator is designed: the problem and goals, the system architecture, the hardware and how it is wired, the control and alarm logic, the software, the mechanical parts, the calibration and test results, and the safe use. It is written so that another engineer can inspect, rebuild or extend the prototype.

## 2. Problem and design goals

Premature and low-birth-weight babies lose heat faster than they can make it, and hypothermia in the first hours is strongly linked to newborn deaths [1]. Certified incubators [3] prevent this, but many smaller hospitals cannot buy or repair them, and their control software cannot be inspected. Syncubator is designed around these goals:

- Keep the baby's surroundings (air temperature, humidity, oxygen) stable at the operator's set points.
- Raise **prioritised alarms on the incubator itself**, so they work with no network and no app running.
- Report a failed sensor as a **fault**, never as a believable wrong reading.
- Use common, replaceable parts and open software that can be inspected and repaired locally.
- Show everything on the bedside touchscreen and, securely, on phones and the web.
- Fail safe: every relay starts off, the phototherapy light is off after any restart, and the lid can only be opened one step.

## 3. System architecture

A Raspberry Pi 4 runs one Python program. Background threads read every sensor into a shared state dictionary. Control loops and the alarm engine act on that state, and a Flask API exposes it. Gunicorn (one worker, eight threads) listens on localhost only; Nginx terminates HTTPS and serves the web app. The Flutter app polls the API once a second and sends commands.

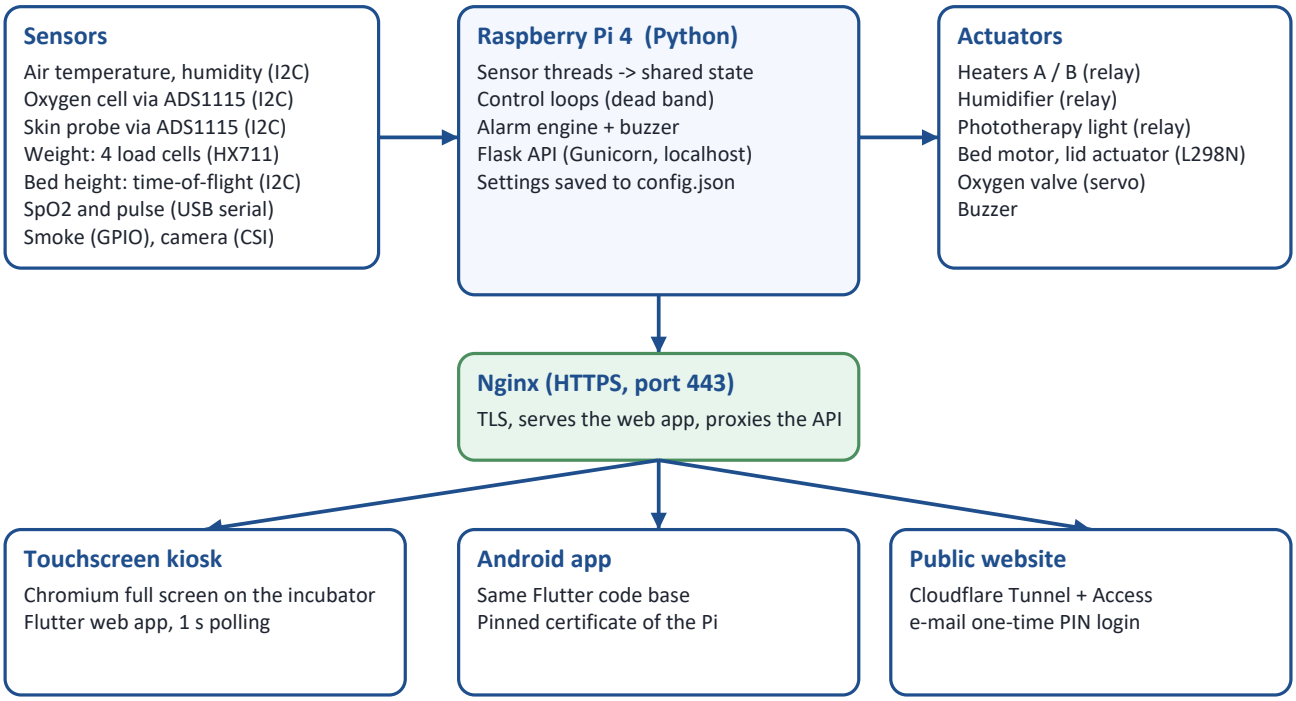



*Figure 1. System architecture: sensors, the Raspberry Pi, actuators and the three ways of reaching the incubator.*

The same web build runs in a full-screen Chromium kiosk on the incubator's touchscreen and on the public website; the Android app is the same Flutter code. The public site *app.syncubator.in* reaches the Pi through a Cloudflare Tunnel, behind Cloudflare Access (e-mail one-time-PIN login). A small Cloudflare Worker shows an "incubator offline" page when the Pi is down. The phone app trusts only the incubator's pinned certificate.

# 4. Hardware

## 4.1 Sensors

| Measurement | Part | Interface | Address / pin (BCM) |
|---|---|---|---|
| Air temperature | 2 x TMP117 (averaged with the BME680) | I2C | 0x48, 0x49 |
| Temperature, humidity, pressure | BME680 | I2C | 0x77 (or 0x76) |
| Humidity (second source) | MS8607 | I2C | 0x40 |
| Oxygen concentration | Electrochemical O2 cell + ADS1115 ADC | I2C, channel A0 | 0x4A |
| Skin temperature | YSI-400 probe, divider into the ADS1115 | I2C, channel A3 | 0x4A |
| Baby weight | 4 load cells with HX711 amplifiers | GPIO | DT 22, 5, 19, 21; SCK 17 |
| Mattress height | VL53L4CD time-of-flight | I2C | 0x29 |
| SpO2 and pulse | Contec CMS60C oximeter, neonatal wrap probe | USB serial, 115200 8N1 | USB |
| Smoke | MQ-2 module, digital output through a divider | GPIO | 9 |
| Video | IMX708 camera (MJPEG stream) | CSI | rpicam-vid |

### 4.2 Actuators

| Function | Driver | Pin (BCM) | Notes |
|---|---|---|---|
| Heater A and heater B | Relay module (active low) | 23, 24 | Redundant pair; one is active, the other is off |
| Humidifier | Relay module | 16 | Same dead-band rule as the heaters |
| Phototherapy light | Relay module, switching the light's return (low side) | 4 | Starts off; state is not saved |
| Bed motor (up / down) | L298N channel A, IN1 / IN2 | 25, 26 | Gated by the time-of-flight sensor, 20 s time-out |
| Lid linear actuator | L298N channel B, IN3 / IN4, ENB held high | 12, 20 | 12 V, 50 mm stroke; timed moves |
| Oxygen valve | Servo (PWM), 8 positions | 18 | Angle = 65 + 10 x level, level 0 to 7 |
| Buzzer | Two-pin buzzer | 27 | Rhythm depends on alarm priority |

All GPIO numbers are BCM numbers. Relays are active low, so every relay pin is set high (off) in the same call that makes it an output; nothing switches on while the program starts. Settings (set points, alarm settings, calibration) are saved to a JSON file written atomically, so they survive a power cut, even one in the middle of a write.

### 4.3 Wiring notes

- **ADS1115** (address 0x4A, ADDR tied to SDA): A0 is the oxygen cell, A3 is the skin probe. VDD is 3.3 V so no input can exceed the supply.
- **Lid actuator:** the L298N's second channel; ENB must be held high. Driven from 12 V through the driver, which lowers the voltage at the actuator by a few volts compared with a direct supply.
- **Phototherapy light:** supply → buck converter OUT+ → light (red wire); the light's black wire goes to the relay COM and the converter OUT- to the relay NO. The relay is wired to the low side and normally open, so the light is off whenever the relay is off.
- **Relay safety:** the relay outputs are rated for the usual 10 A relays; every mains load needs its own fuse and an enclosure that cannot be touched.
- **Smoke sensor:** the module's DO output swings to 5 V, so it goes through a resistor divider to GPIO 9. A wire that comes off pulls the pin low and raises the alarm, so a broken wire is never silent.

## 5. Control design

### 5.1 Temperature and humidity

Air temperature is the average of the two TMP117 sensors and the BME680; humidity is the average of the BME680 and MS8607. A failed sensor is dropped from the average. The heaters and humidifier are relays, so a plain threshold would chatter at the set point. A dead band is used instead (the loop runs every 2 s):

> heater ON when air temperature < set point − 1.5 °C; heater OFF when air temperature > set point + 1.5 °C

Humidity uses the same rule. Heater B is a spare. Switching control off forces every relay off.

### 5.2 Oxygen

The oxygen valve is a servo with eight positions (level 0 closed to level 7 fully open). The control loop moves it only to fully open or fully closed, with a 1.5-percentage-point dead band around the target, so the valve does not chatter. The oxygen reading comes from the cell through a two-point calibration (0 V is 0 %, 1.4 V is 20.9 %).

### 5.3 Bed and lid

- **Bed:** the time-of-flight sensor gates the motor. Raise is allowed from 8 to 15 cm (target 19.2 cm); lower from 15 to 25 cm (target 12 cm). A move stops after 20 s at the latest.
- **Lid:** there is no position sensor, so moves are timed. Open retracts the actuator for 6 s; Close extends it for 6 s. Open is refused until Close has been used, so the lid can never be opened more than one step. Any command is

refused while the lid is moving, and the position survives a restart.

- **Phototherapy light:** an on/off switch in the app. It is deliberately off after every restart.

# 6. Skin temperature measurement

The YSI-400 probe is a thermistor of 2252 Ω at 25 °C. It is read as one half of a divider into the ADS1115:

3.3 V → R_ref (five 2.2 kΩ in series, about 11 kΩ) → A3 ← probe → GND

The probe's resistance follows from the voltage V at A3 and the supply $V_{dd}$:

$R_{probe} = R_{ref} \times V / (V_{dd} - V)$

It is converted to temperature with the Steinhart-Hart equation, $1/T = A + B \cdot \ln R + C \cdot (\ln R)^3$, with $A = 1.468 \times 10^{-3}$, $B = 2.383 \times 10^{-4}$, $C = 1.007 \times 10^{-7}$ (T in kelvin). This reproduces the probe's table: 2252 Ω gives 25.0 °C, 1355 Ω gives 37.0 °C and 7355 Ω gives 0 °C. The program takes the median of eight conversions each second. The expected voltage at A3 is about 0.56 V at 25 °C and about 0.36 V at 37 °C.

| Status | Meaning | Rule |
|---|---|---|
| OK | Valid reading | Temperature between 5 and 60 °C |
| PROBE_OFF | Nothing in the jack | Voltage at or above 97 % of the supply |
| SHORT | Probe lead shorted | Resistance below 150 Ω |
| RANGE | Not a skin temperature | Outside 5 to 60 °C |
| NOT_CONNECTED | ADS1115 not answering | I2C error |

# 7. Alarm system

The alarm engine runs on the Pi, not in the app, because an alarm in a client goes quiet exactly when it matters: when the browser crashes or the Wi-Fi drops. It checks every condition once a second and uses the low, medium and high priorities of IEC 60601-1-8 [2].

- **Onset delay:** a condition must hold for 10 s before it fires (1 s for smoke), so one noisy sample cannot raise an alarm.
- **Latching:** an alarm that clears by itself stays on screen until it is acknowledged.
- **Acknowledge:** silences the buzzer for 120 s. If the problem is still there, the buzzer sounds again.
- **Buzzer rhythm by priority:** high 0.15 s on / 0.15 s off; medium 0.30 s on / 0.70 s off; low 0.25 s on / 3.0 s off.
- **Sensor faults:** if a sensor dies, the alarms that depend on it give way to one sensor-fault alarm, so a zero reading cannot raise a false "temperature low".

| Priority | Alarm | Raised when |
|---|---|---|
| High | Heater runaway | Air temperature is far above the set point with control on |
| High | Air temperature low / high | Air temperature leaves the set range |
| High | Oxygen low / high | Oxygen concentration leaves the set range |
| High | Smoke | Smoke sensor triggered (or its wire off) |
| High | Skin temperature low / high | Skin temperature leaves the set range |
| High | SpO2 low; pulse low / high | Reading leaves the set range |
| Medium | SpO2 high | Reading above the set range |
| Medium | Humidity low / high | Humidity leaves the set range |
| Medium | SpO2 probe off; skin probe off or faulty | No valid reading, while monitoring is on |
| Medium | Climate or oxygen sensor not responding | No reading returned |

| Priority | Alarm | Raised when |
|---|---|---|
| Low | Bed-distance or weight sensor not responding | Sensor absent or not OK |

The SpO2 and skin-temperature alarms only run while the operator has switched monitoring on for that card, so an empty probe does not raise an alarm. A separate SOS call sounds the buzzer continuously and shows a banner on every client until it is cancelled.

# 8. Software design

## 8.1 Backend (Raspberry Pi)

*app.py* contains the hardware drivers, ten background threads (weight, climate, bed, oxygen, alarms, buzzer, SpO2, scale start-up, smoke, skin probe), the alarm engine and the API. All threads share one state dictionary guarded by a lock; the ADS1115 has its own lock because the oxygen and skin readings use the same multiplexer. The service runs under systemd and starts on boot.

| Endpoint | Method | Purpose |
|---|---|---|
| /status | GET | Everything: climate, oxygen, weight, bed, lid, SpO2, skin, smoke, light, alarms |
| /climate/settings | POST | Set point, humidity target, control on or off, heater choice |
| /oxygen/level, /oxygen/settings | POST | Valve level; target and automatic control |
| /bed/move | POST | Raise or lower the mattress (gated by the distance sensor) |
| /lid/move | POST | Open or close the lid (timed, with the one-step rule) |
| /photo | POST | Phototherapy light on or off |
| /weight/tare, /weight/restart | POST | Zero or restart the scales |
| /spo2/settings, /skin/settings | POST | Switch SpO2 or skin alarms on or off |
| /alarms, /alarms/acknowledge | GET, POST | Alarm detail and silence |
| /sos | POST | Raise or cancel the SOS call |
| /video_feed | GET | Live camera stream (MJPEG) |
| /*/restart | POST | Re-initialise one sensor driver without restarting the service |

## 8.2 Application (Flutter)

The app is written once in Flutter [9] and uses a Provider + Service + Model structure: a model per subsystem, a service for the HTTP calls, and a provider that polls once a second. Each subsystem is a card on the home screen: baby information, climate, SpO2 and pulse, skin temperature, oxygen, weight, bed, lid and phototherapy light. A banner at the top shows the most serious alarm and acknowledges it on a tap. A demo mode with simulated data lets the interface be tested without hardware. Automated widget tests cover the lid, skin and phototherapy cards.

- The kiosk passes taps but not drags on the touchscreen, so scrolling uses arrow buttons.
- The alarm tone comes from the Pi's buzzer, because Chromium blocks audio until the first tap.
- Each sensor card can restart its own driver; a hidden long press leaves the kiosk for the Pi desktop.

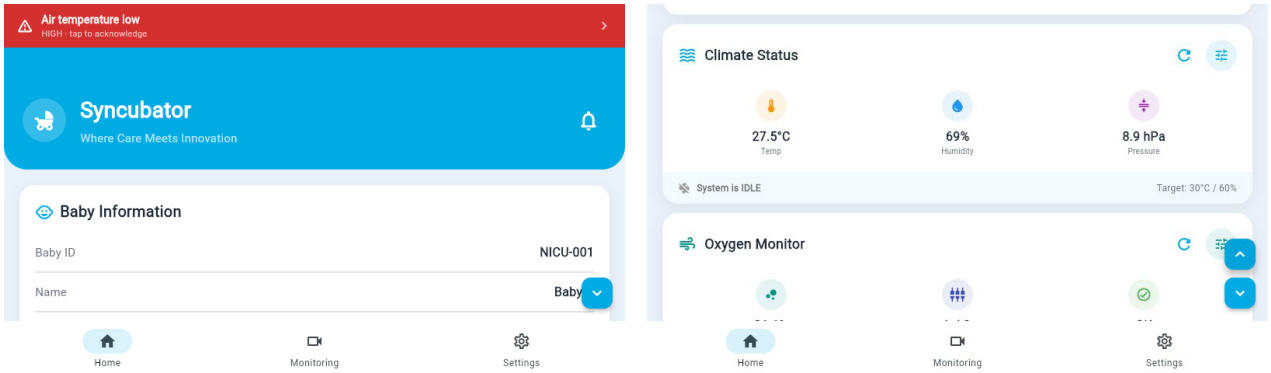



*Figure 2. The application: a high-priority alarm banner (left) and the climate and oxygen cards (right).*

# 9. Mechanical design

The enclosure, canopy and mattress platform were modelled in SolidWorks. A white 3D-printed canopy with acrylic panels sits on an aluminium frame; the oxygen cylinder, regulator and a drip-pole style stand are on the side.

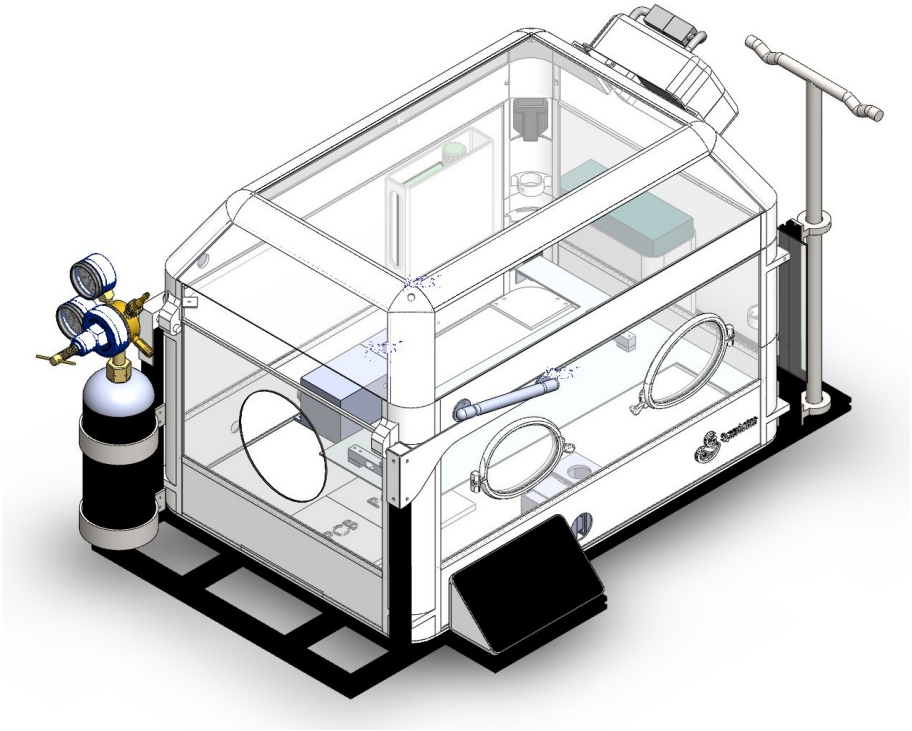


*Figure 3. CAD model of the incubator with the canopy closed.*

## 9.1 Lid and actuator

The canopy is a lid hinged on its rear rail. A 12 V linear actuator (50 mm stroke) pushes it through a travel of about 58 degrees. Its upper end is held by a pair of L-brackets on the lid bar; its lower end is clamped to a vertical strip that stands on a corner bracket fixed to the front face of the rail, with the motor behind the tube. The assembly was checked for interference through the whole sweep. A two-piece 3D-printed capsule cover, with a fixed skirt down to the rail, covers the actuator.

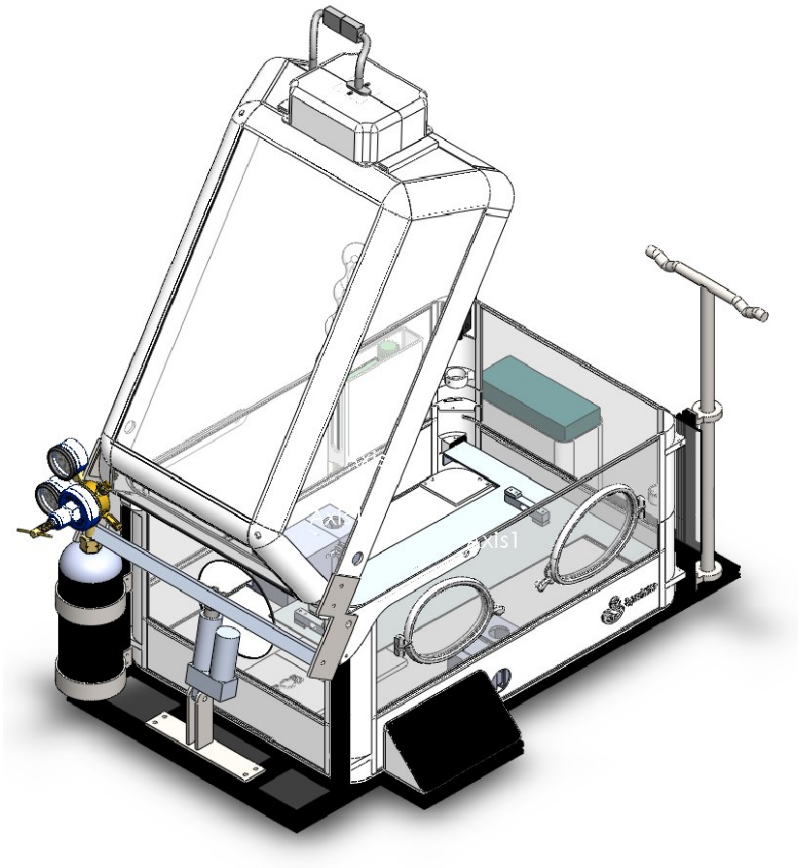


*Figure 4. The lid raised on its actuator (CAD).*

## 9.2 Access door and hand hole

The front access door has one large elliptical hand hole, 360 x 130 mm, closed by a drop-down flap hinged at its bottom edge. The flap stops at 90 degrees (horizontal) and latches at the top with a turn button. The ring halves, hinge parts and latch are 3D printed; the panel and the 2 mm flap are cut from acrylic (DXF files are provided). The flap is clear in both the closed and the open position.

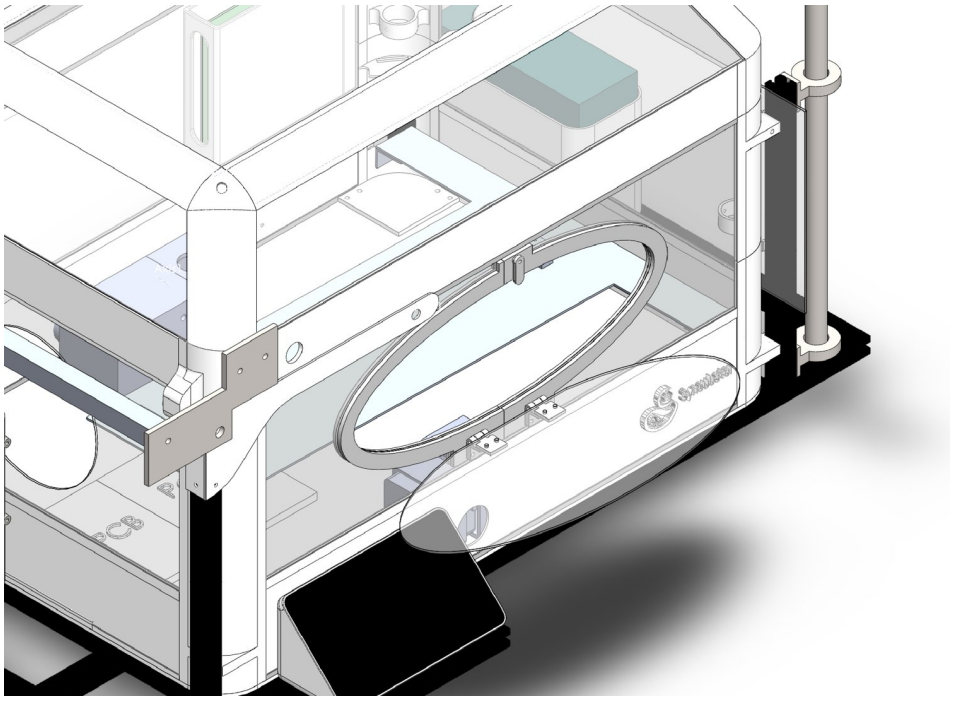


*Figure 5. The access door with the elliptical hand hole and its flap (CAD).*

# 10. Calibration, testing and results

| Item | What was done | Result |
|---|---|---|
| Load cells | Zeroed on the empty platform, scaled against a 5 kg mass | Old constants read 12.84 times too low; the platform now reads within a few tens of grams of zero |
| Oxygen cell | Two-point calibration: 0 V is 0 %, 1.4 V is 20.9 % | Reads 21.0 to 21.1 % in room air |
| Alarm timing | Breaches held for 5 s and for longer on the hardware | 5 s stayed silent; a sustained breach fired at 10 s; acknowledging silenced the buzzer and kept the alarm listed |
| Smoke alarm | Module output driven low on the bench | The alarm fired after its 1 s hold |
| Lid | Open and close from the kiosk | Each move ran 6 s; Open stayed locked until Close was used |
| Skin probe curve | Checked the equation against the YSI-400 table | 25.0, 37.0 and 0.0 °C at 2252, 1355 and 7355 Ω |

Bring-up exposed four faults, all fixed: a disconnected load cell hung the service until a 60 s worker time-out was added; the camera went black on reconnect until the old camera process was stopped first; new endpoints were swallowed by a stale Nginx path list; and settings were lost on restart until atomic saving was added.

# 11. Safety and use

- **Research prototype.** Syncubator is a working research prototype and is not a certified medical device. Any clinical use needs regulatory approval and testing to IEC 60601-1-8 and 60601-2-19 [2][3].
- **Fail-safe by design:** every relay starts off, the phototherapy light is off after any restart, the lid can only be opened one step, and the alarm engine and buzzer work with no network.
- **Secure access:** remote access sits behind an e-mail one-time-PIN login and HTTPS, and the phone app trusts only the incubator's pinned certificate.

# References


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|---|---|
| [1] | WHO, Thermal Protection of the Newborn, 1997. |
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